Arista Access Node WFD-ACC-59 QoS Congestion Event SOP (ALARM_9983)

Node: WFD-ACC-59 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9983

Description

This Standard Operating Procedure (SOP) addresses a Major Severity QoS Congestion Event on the Arista Access Node WFD-ACC-59 located at the Whitefield POP. The alarm code ALARM_9983 indicates a potential Microburst, interface tail drop, or QoS queue starvation issue within the access node's QoS domain. This event is likely caused by an imbalance in traffic distribution, inadequate QoS policing, or insufficient QoS buffer allocation, leading to a degradation in network performance. The Arista EOS (Extensible Operating System) relies on the Diffserv (Differentiated Services) architecture to classify and prioritize traffic, ensuring that critical traffic flows are delivered with the required Quality of Service (QoS). However, in the event of a congestion event, the access node's QoS queues may become starved, causing packet loss and subsequent network instability.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that the remote hands are dispatched.

- Check the redundant link status to ensure that there are no issues with the backup connections.

- Confirm that the network monitoring tools are functioning correctly and providing accurate data on network performance.

Potential Root Causes

Cause	Probability
Incorrect QoS configuration	High
Insufficient QoS buffer allocation	Medium
Microburst or interface tail drop	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos statistics, Expected Output: QoS statistics for interface Ethernet1/1: packets output: 123456789 packets dropped: 123456 , Description: This command provides an overview of the QoS statistics for the specified interface, including packets output and dropped, to help isolate the source of the congestion event.

Command: show interface Ethernet1/1 detail, Expected Output: Ethernet1/1 is up, line protocol is up Hardware is i40e 100G 8x 10GbE SFP+, address is 00:11:22:33:44:55 (bia 00:11:22:33:44:55) MTU 1500 bytes, BW 10000000 Kbit, DLY 100 usec Rcvd: 123456789 packets (321321321 bytes) 123456 packets received with intra-cell errors , Description: This command provides detailed information about the interface, including the current state, statistics, and any errors
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that may be contributing to the congestion event.

Command: show qos queue statistics, Expected Output: QoS queue
statistics for interface Ethernet1/1: queue 1: packets output:
123456789 packets dropped: 123456 , Description: This command provides
a detailed view of the QoS queue statistics for the specified
interface, including packets output and dropped, to help isolate the
source of the congestion event.
```

Resolution Steps

Step 1: Step: Verify QoS configuration, Command: show qos configuration, Description: Verify that the QoS configuration is correct and that the policies are properly applied to the interface. Check for any misconfigurations or inconsistencies that may be contributing to the congestion event.

Step 2: Step: Adjust QoS buffer allocation, Command: config qos buffer allocation adjust, Description: Adjust the QoS buffer allocation to ensure that there are sufficient buffers to handle the traffic load. This may involve increasing the buffer size or adding additional buffers to the interface.

Step 3: Step: Implement traffic policing, Command: config qos policing, Description: Implement traffic policing to prevent excessive traffic from overwhelming the interface. This may involve configuring policers to limit the rate of traffic or to discard packets that exceed the policer threshold.

Post-Check Verification

```
Command: show qos statistics, Expected Output: QoS statistics for
interface Ethernet1/1: packets output: 123456789 packets dropped: 0 ,
Description: Verify that the QoS statistics show a significant
reduction in packet drops and a corresponding increase in packets
output, indicating that the resolution steps were effective.

Command: show interface Ethernet1/1 detail, Expected Output:
Ethernet1/1 is up, line protocol is up Hardware is i40e 100G 8x 10GbE
SFP+, address is 00:11:22:33:44:55 (bia 00:11:22:33:44:55) MTU 1500
bytes, BW 10000000 Kbit, DLY 100 usec Rcvd: 123456789 packets
(321321321 bytes) 0 packets received with intra-cell errors ,
Description: Verify that the interface statistics show a significant
reduction in errors and a corresponding increase in packets received,
indicating that the resolution steps were effective.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: config qos configuration restore, Description: Restore the original QoS configuration to ensure that the interface returns to its previous state. This may involve restoring the original policer settings or buffer allocation.

- Step: Verify QoS configuration, Command: show qos configuration, Description: Verify that the QoS configuration is restored to its original state, ensuring that the interface is functioning correctly.